

PC5203 Advanced Solid State Physics

Weeks 10-11, due Tuesday 6 April 2021

[main concepts to cover: Contour ordered Green's function. Boltzmann equations. SSH model, Berry phases.]

1. Let G be a contour ordered Green's function $G_{jk}(\tau, \tau')$ with components $G^{\sigma\sigma'}$ satisfying the relation $G^t + G^{\bar{t}} = G^< + G^>$. We define the inverse of G as $B = G^{-1}$ by the equation $\int d\tau'' G(\tau, \tau'') B(\tau'', \tau') = I \delta(\tau, \tau')$. Here I is the identity matrix and δ is the delta function on contour. (a) Show explicitly we also have the relation $B^t + B^{\bar{t}} = B^< + B^>$ (Hint: go into energy E space so that convolution in time becomes simple matrix multiplication, or use Keldysh rotation. We also assume the inverse of matrices exist whenever it is needed). (b) Show that the Dyson equation on contour, $G = g + g\Sigma G$ is equivalent to $G^{-1} = g^{-1} - \Sigma$. (c) From this, show that the self energy Σ also satisfies the same constraint, i.e., $\Sigma^t + \Sigma^{\bar{t}} = \Sigma^< + \Sigma^>$.
2. Consider the electron-electron Coulomb interaction of the form $H' = \frac{1}{2} \sum_{jl} c_j^\dagger c_l^\dagger v_{jl} c_l c_j$ on a simple cubic lattice, so that the site j or l is defined on the lattice site on a system of $N = L^3$ sites. Transform the Coulomb interaction into $\mathbf{k}$ space, by $c_j = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} \tilde{c}_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{R}_j}$, assuming v is translationally invariant, $v_{jl} = v(\mathbf{R}_j - \mathbf{R}_l)$, determine the interaction in $\mathbf{k}$ space. Here $\mathbf{R}_j$ is the real space lattice vector for the site j and $\mathbf{k}$ varies over the first Brillouin zone of the simple cubic lattice. (b) Using H' as the small interaction in the Fermi golden rule, determine the form of the collision rate $\left(\frac{\partial f}{\partial t}\right)_{\text{colli}}$ for the Boltzmann equation under Coulomb scattering potential. [Hint: this is formally very similar to the original Boltzmann equation for classical particles, except that we have to take the Pauli exclusion principle into account for the fermionic electrons]. (c) Show that the collision rate is zero when the distribution f is given by the equilibrium Fermi distribution $f^0 = 1/(e^{\beta(\epsilon - \mu)} + 1)$.
3. Consider the Rice-Mele model $\hat{H} = c^\dagger H c$ with a Hamiltonian which is the one-dimensional Su-Schrieffer-Heeger (SSH) model with an extra onsite potential of ϵ or $-\epsilon$ depending on if the site label is even numbered or odd numbered (that is, $\epsilon c_j^\dagger c_j$ if j is even, or $-\epsilon c_j^\dagger c_j$ if j is odd). The hopping parameter is t_1 between sites in the same unit

cell of lattice constant $2a$, and t_2 between neighboring unit cells. (a) Show that the Hamiltonian in k space is

$$H(k) = \begin{bmatrix} \epsilon & -t_1 - t_2 e^{-2ika} \\ -t_1 - t_2 e^{+2ika} & -\epsilon \end{bmatrix}.$$

(b) Determine the eigenvalues and eigenvectors of the Hamiltonian $H(k)$. (c) Compute the Zak phase (i.e. Berry phase in 1D k -space) for the lower band and make a plot of the Zak phase as a function of t_2 and ϵ in a 2D density plot, fixing $t_1 = 1$.